

1 Saturn's Hexagon

Parallel shallow-water modelling on a rotating sphere

1.1 Jet speed and polar structure

Exercise 1. Where do cyclones and anticyclones form relative to the eastward polar jet?

Model answer. The two signs of vorticity develop on opposite sides of the jet. For a predominantly zonal eastward flow $u(y)$, the leading contribution to relative vorticity is

$$\zeta \simeq -\frac{\partial u}{\partial y}.$$

On the poleward side of a jet maximum, u decreases towards the pole, so $\partial u / \partial y < 0$ and $\zeta > 0$: cyclonic vorticity is favoured. On the equatorward side, u increases towards the jet maximum, so $\partial u / \partial y > 0$ and $\zeta < 0$: anticyclonic vorticity is favoured. In the simulation the resulting vortical structures therefore organise on opposite flanks of the jet rather than appearing randomly across the polar cap.

Exercise 2. What happens to the simulated flow as the maximum jet speed is increased from 60 to 420 m s⁻¹?

Model answer. Increasing u_{jet} strengthens the horizontal shear and makes the jet increasingly susceptible to large-amplitude meanders and vortical deformation. The response is not simply “a faster jet gives a better hexagon”. At 60 m s⁻¹ the polar boundary remains comparatively weakly distorted and close to axisymmetric. Around 90–120 m s⁻¹ a clear polygonal boundary develops, with a hexagon-like form particularly evident. At 200 m s⁻¹ the deformation is stronger and more nonlinear. By 420 m s⁻¹ the flow is highly distorted and lopsided, and the clean hexagonal appearance is degraded. The experiment therefore illustrates a range of instability strength: too little shear gives little polygonal structure, while very strong shear produces a more chaotic nonlinear state.

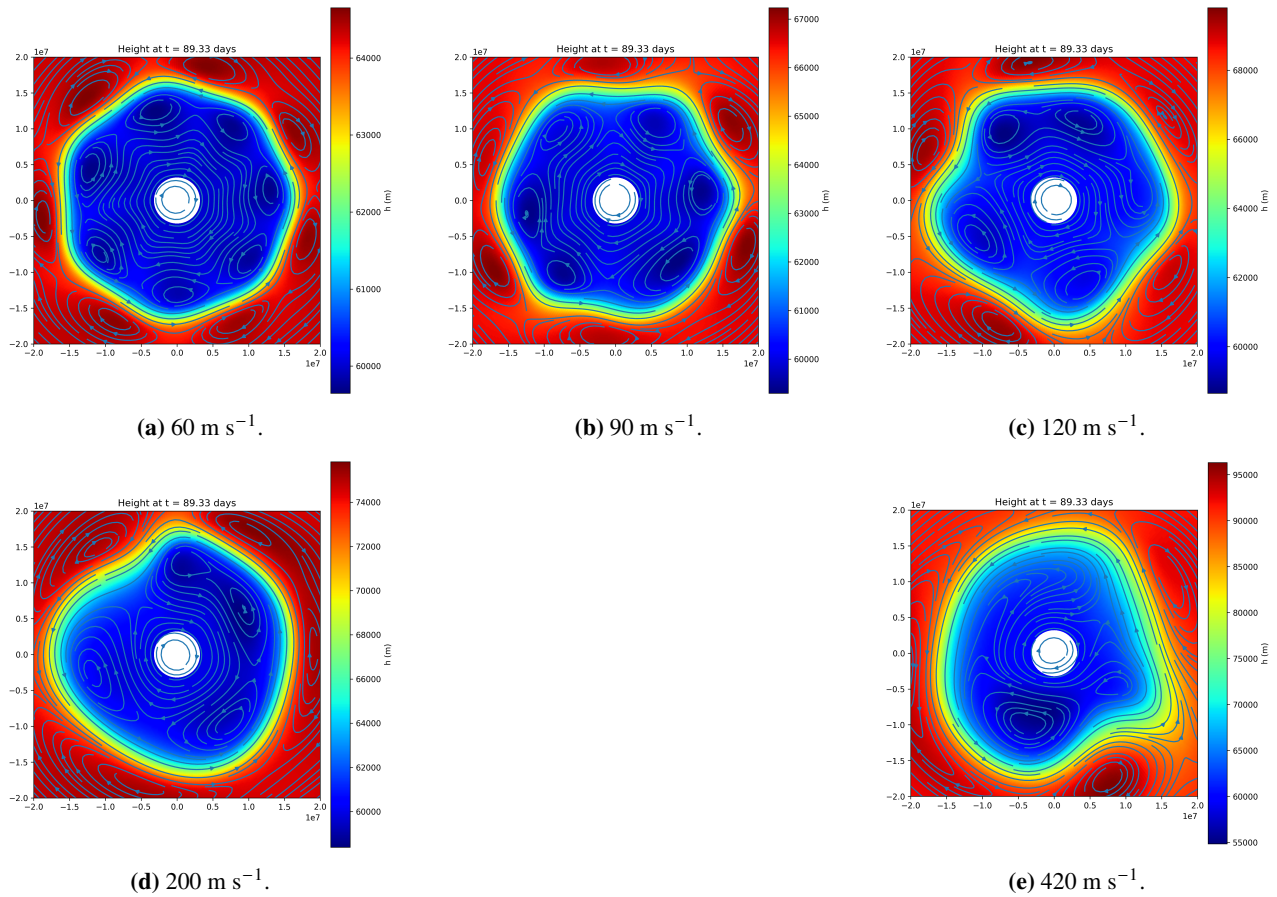


Figure 1: Final-state images for the prescribed jet-speed sequence. Polygonal structure is clearest at intermediate jet speeds; the strongest jet becomes much more distorted.

Overall interpretation

Model answer. The practical demonstrates that Saturn-like polygonal structure can emerge from the dynamics of a rapidly rotating, strongly sheared polar jet without prescribing a polygonal boundary directly. The exact morphology depends on the jet profile, rotation, layer properties, numerical setup and integration time, so the simulation should be interpreted as an idealised dynamical analogue rather than a unique reconstruction of Saturn's atmosphere. The useful result is the systematic comparison: changing only the jet speed changes the strength and organisation of the instability and therefore the shape of the polar flow.